Federated Learning for Hospital Readmission Prediction

Authors: Sara Haile and Luca Bianchi (fictional authors; synthetic evaluation paper).

Identifier: doi:10.5555/ras-eval.p09 (fictional, not registered).

Abstract

Seven hospitals trained a readmission model without sharing patient records.

The federated model reached an AUROC of 0.78, close to 0.79 for pooled training.

Introduction
Readmission within 30 days is costly for hospitals and harmful for patients.
Prediction models need data from many hospitals to generalise across populations.
Legal rules often prevent hospitals from pooling patient records in one place.
Federated learning trains a shared model while each hospital keeps its own data.
We test whether federated training can match a model trained on pooled records.
We also measure the cost of adding differential privacy to the shared updates.
Our cohort comes from seven hospitals of very different sizes in one region.
All analyses were approved by the ethics boards of the participating hospitals.

Methods

Each hospital ran 5 local epochs per round for 60 rounds of federated training.
Updates were clipped to norm 1.0 and aggregated with secure summation.
The model is a gradient-boosted ensemble distilled into a small neural network.
Features include diagnoses, length of stay, age group and prior admissions.
The outcome is an unplanned readmission within 30 days of hospital discharge.
Pooled training on the same features serves as an upper reference model.
Each hospital held out 20% of its admissions for the final evaluation step.
We repeated federated training with three random seeds for every setting.

Results

Per-hospital AUROC values are listed in Table 4, which is provided as a scanned image.

The smallest hospital, with 1,900 admissions, gained the most from federation.

Differential privacy with epsilon 8 reduced the federated AUROC to 0.75.

Calibration of the federated model was similar to that of the pooled model.

Training finished in 11 hours of wall-clock time across all seven hospitals.

Communication per round stayed below 4 megabytes for every participating site.

Hospitals with fewer than 5,000 admissions benefited more than large hospitals.

No hospital performed worse with the federated model than with a local model.

Discussion

Federated training nearly matched pooled training without moving patient records.

The privacy cost of epsilon 8 may be acceptable for decision support but not for all uses.

Small hospitals gain the most, which supports regional collaboration on such models.

Our cohort covers one region, so external validation in other regions is needed.

Scanned appendix tables should be converted to text before any automated reuse.

Future work will test personalisation layers for hospitals with unusual case mixes.

Secure summation added little overhead compared with the local training time.

Clinicians reviewed the top features and found them clinically plausible.

Conclusion

Federated learning offers near-pooled readmission prediction across seven hospitals. Per-hospital values remain available only in the scanned Table 4 of the original.

References

[1] E. Novak. Secure Aggregation for Federated Health Models. 2024.

[2] D. Ferreira. Readmission Risk Scores Revisited. 2023.